Hardening a Network Benchmarking Laboratory

Comparative Analysis of TCP and QUIC Congestion Control

Eyobed

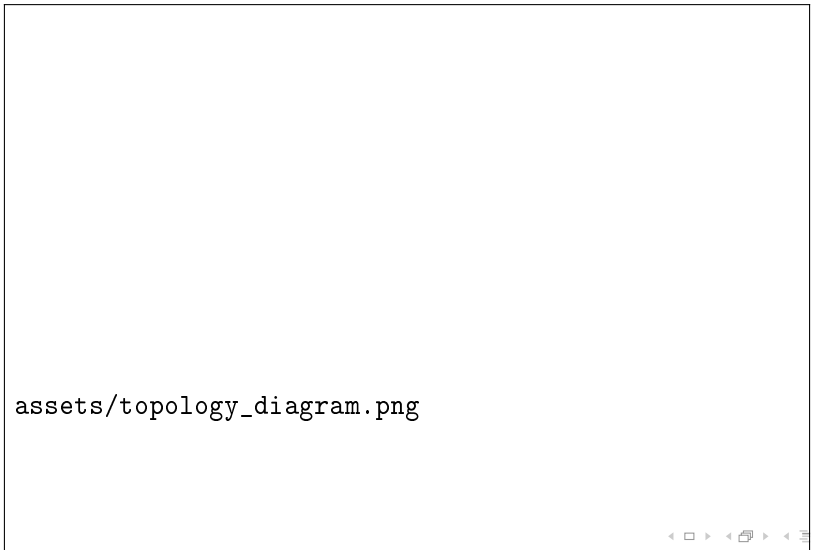
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Experimental Objective

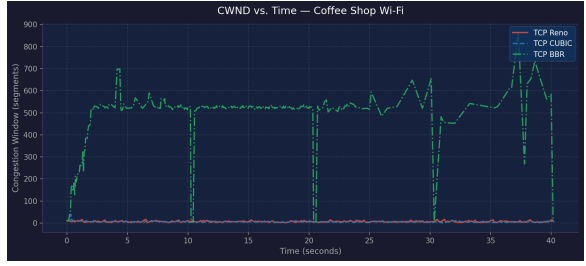
- **Goal:** Evaluate transport protocol efficiency (TCP vs QUIC) in hostile network conditions.
- **Isolation:** Use Linux Network Namespaces to simulate isolated nodes.
- **Rigor:** Moving bottlenecks to the router egress to eliminate measurement artifacts.
- **Telemetry:** High-fidelity 50Hz polling of the Congestion Window (CWND).

Hardened Lab Topology



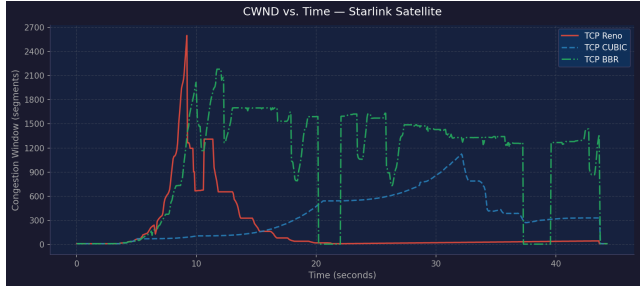
Scenario 1: Coffee Shop Wi-Fi (3% Loss)

- **Condition:** 100Mbps cap + 3% random packet loss.
- **Reno/CUBIC:** Severe collapse due to false congestion identification.
- **BBR:** Immunity to random loss; maintains model-based throughput.



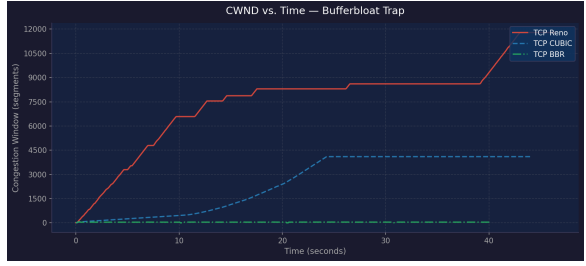
Scenario 2: Starlink Satellite (600ms RTT + Jitter)

- **Observation:** Long-tail ramping periods.
- **BBR Behavior:** The 10-second `ProbeRTT` window becomes visible.
- **Jitter:** Reno fails due to retransmission timeouts triggered by latency variance.



Scenario 3: The Bufferbloat Trap

- **Setup:** 10Mbps cap with a 50,000 packet queue.
- **CUBIC:** Meaningless scaling into the buffer; RTT grows to seconds.
- **BBR:** Locks at BDP, ignoring the bloated queue to preserve latency.



Comparative Summary

Algorithm	WiFi	Satellite	Bloat
TCP Reno	Poor	Failed	Bloated
TCP CUBIC	Fair	Poor	Bloated
TCP BBR	**Excellent**	**Good**	**Target**
QUIC	TBD	TBD	TBD